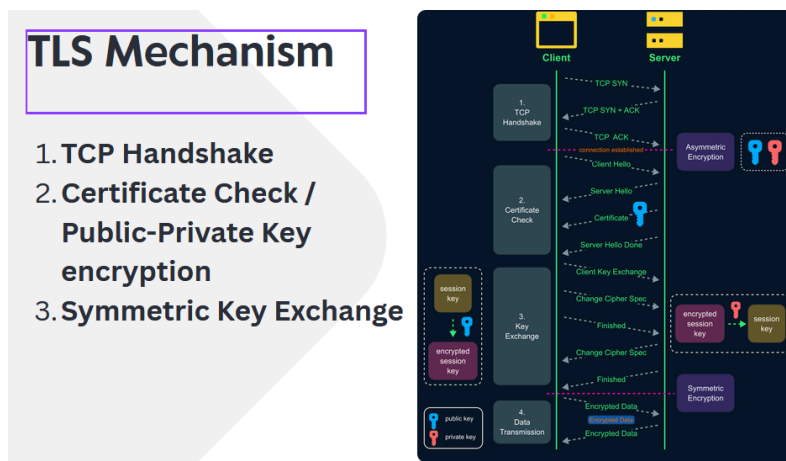


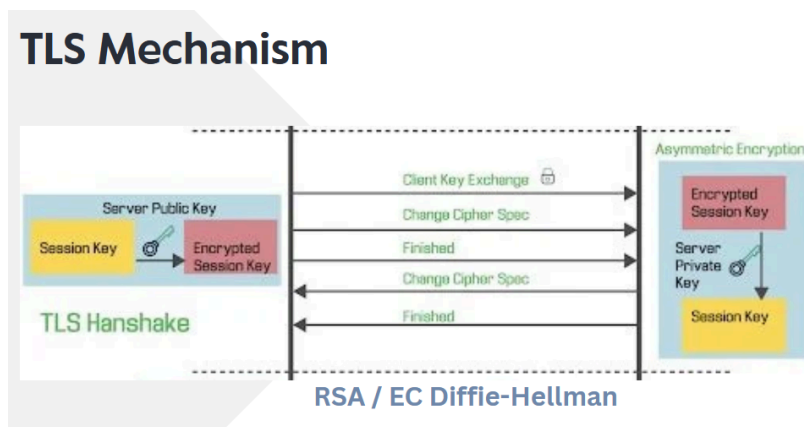
Quantum Key Distribution

Introduction

The Transport Layer Security (TLS) protocol forms the backbone of secure communication on the internet, providing confidentiality, integrity, and authentication for data exchanged between clients and servers. A TLS session is established through a multi-phase handshake that begins after a standard TCP connection is formed. During this handshake, the communicating parties negotiate cryptographic parameters, authenticate each other's identities through digital certificates, and establish a shared secret from which symmetric session keys are derived.

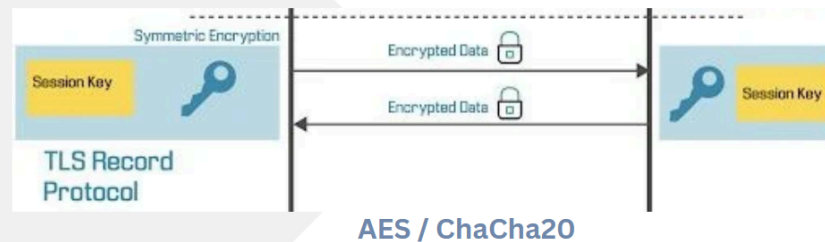


Central to the security of this process are two distinct cryptographic mechanisms. The first is asymmetric encryption, which underpins both certificate verification and the key exchange phase. In widely deployed configurations, the Diffie-Hellman (DH) key exchange protocol or RSA allows two parties to jointly construct a shared secret over a public channel without ever transmitting that secret directly.



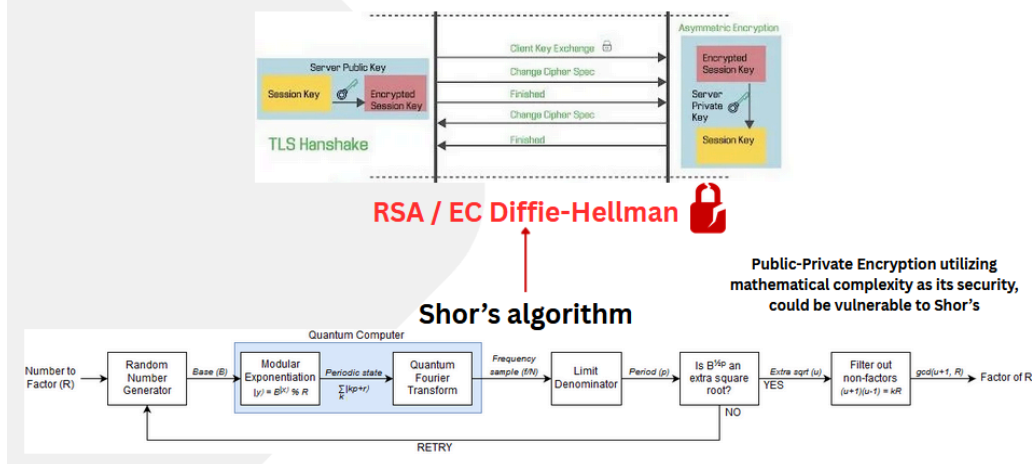
The security of this scheme rests on the computational hardness of the discrete logarithm problem.

TLS Mechanism



The second mechanism is symmetric encryption, where the shared secret derived from the key exchange is used to instantiate a cipher, most commonly the Advanced Encryption Standard (AES) for the bulk encryption of session data. Symmetric encryption is preferred for this role due to its computational efficiency relative to asymmetric schemes.

TLS Vulnerability

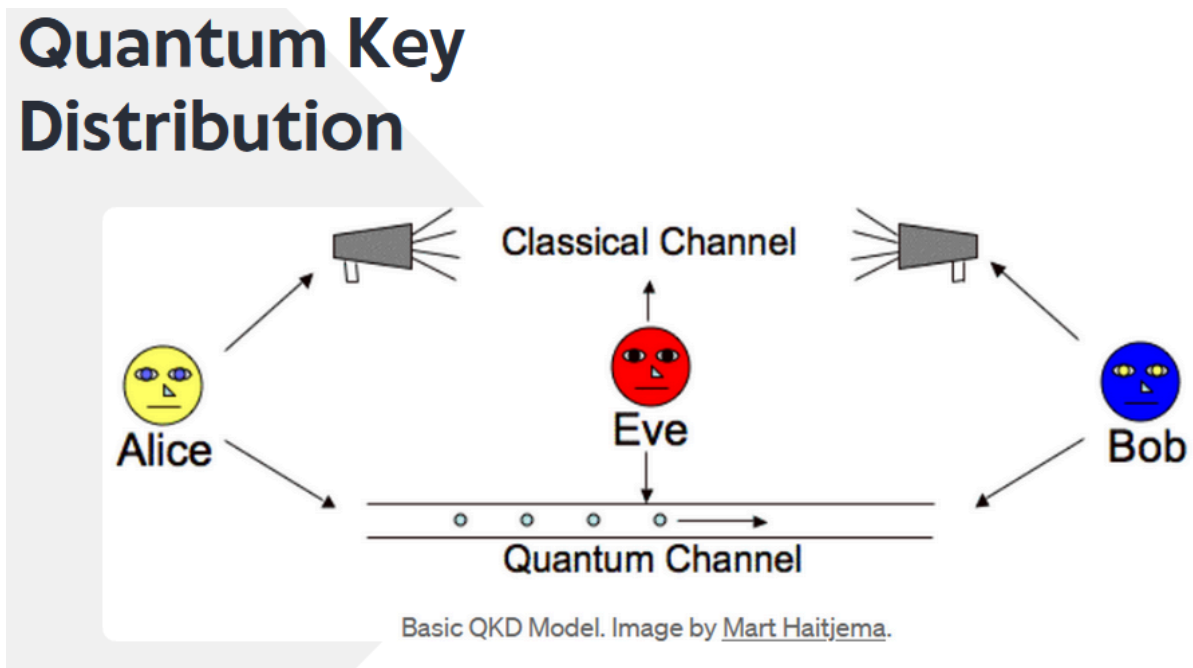


This classical architecture, however, faces an existential threat from quantum computing. Shor's algorithm, if executed on a sufficiently powerful fault-tolerant quantum computer, can solve the discrete logarithm and integer factorization problems in polynomial time, rendering both Diffie-Hellman key exchange and RSA-based certificate schemes cryptographically broken.

While AES retains meaningful security under quantum attack Grover's algorithm provides only a quadratic speedup, effectively halving the key length, the asymmetric components of TLS are fundamentally compromised.

One proposed remedy is Quantum Key Distribution (QKD), which replaces the classical key exchange phase with a protocol whose security is guaranteed not by computational hardness assumptions but by the laws of quantum mechanics. Rather than relying on the difficulty of a mathematical problem that a quantum adversary could solve, QKD exploits fundamental quantum properties such as the no-cloning theorem and the disturbance induced by measurement to detect any eavesdropping attempt and ensure that the shared key established between two parties is information-theoretically secure. Integrating QKD into the TLS handshake therefore addresses the precise vulnerability that Shor's algorithm exposes, while leaving the symmetric AES layer intact.

QKD (BB84)



Quantum Key Distribution (QKD) allows two parties, Alice and Bob, to establish a shared secret key whose security is guaranteed by the laws of quantum mechanics rather than computational hardness. The protocol operates across two channels:

- a quantum channel for transmitting quantum states

- a classical authenticated channel for reconciliation.

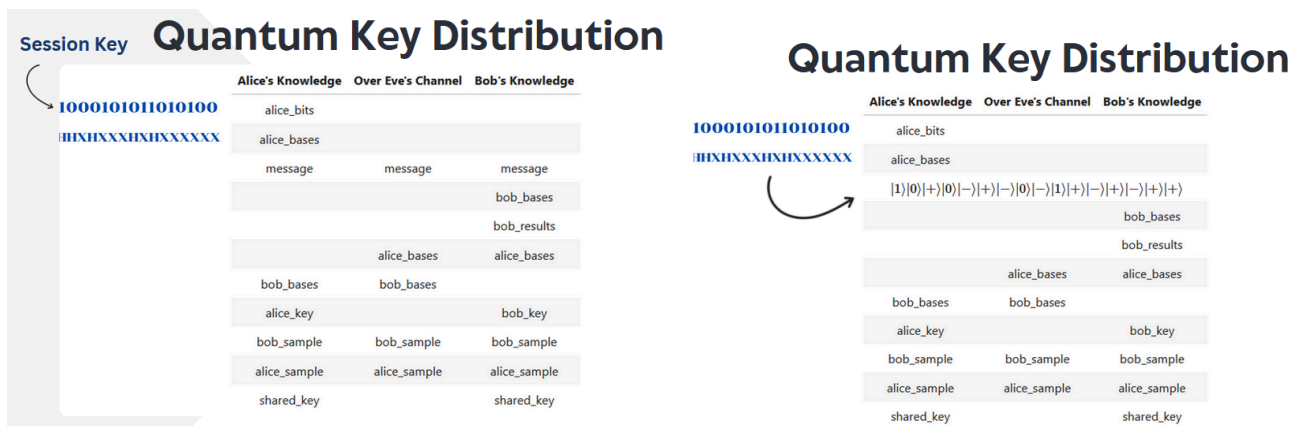
(The classical channel may be observed freely by an adversary, it need not be confidential.)

The core principle is that any measurement on an unknown quantum state disturbs it. An eavesdropper Eve cannot intercept Alice's transmissions without introducing detectable disturbances. After the quantum transmission phase, Alice and Bob use the classical channel to compare a subset of their measurement outcomes and estimate the error rate, known as the Quantum Bit Error Rate (QBER). A low QBER confirms the channel is clean; a high QBER indicates eavesdropping and the session is aborted.

The surviving key bits are then processed through information reconciliation, which corrects residual errors from channel noise, and privacy amplification, which compresses the key to eliminate any partial information Eve may have acquired. The output is a shared secret key that is information-theoretically secure, unconditionally safe against any adversary regardless of computational power, including a quantum computer running Shor's algorithm.

QKD (BB84)

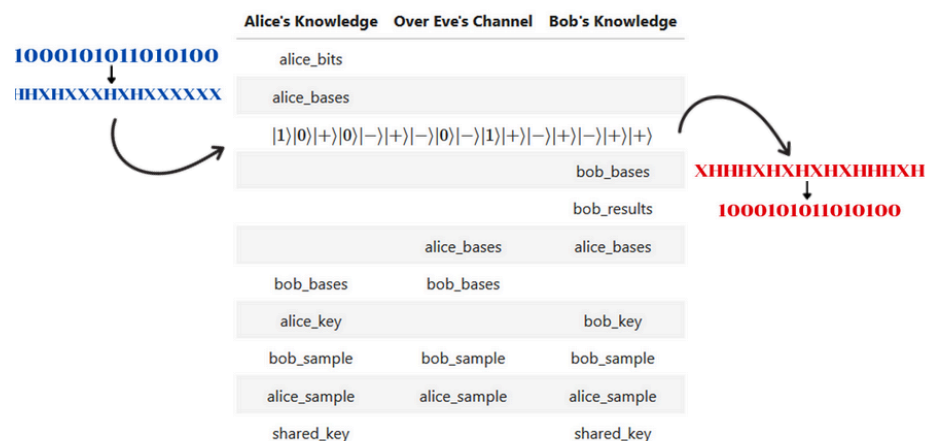
BB84, proposed by Bennett and Brassard in 1984, is the foundational QKD protocol. It encodes classical bits into quantum states using two incompatible measurement bases the Z basis and the X basis such that any eavesdropping attempt is detectable.



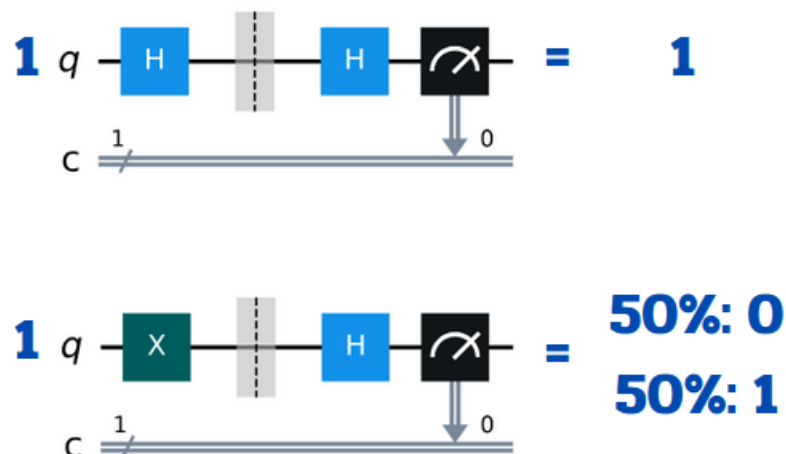
Key Generation and Encoding. Alice generates two independent random bit strings of length n . The first is her raw key, the classical bits she wishes to encode. The second determines her choice of basis for each bit Z or X. For each bit, Alice prepares a qubit as follows: in the Z basis she applies an X gate to encode a 1 (leaving the qubit as $|0\rangle$ for a 0 and $|1\rangle$ for a 1), and in the X

basis she applies a Hadamard gate to encode into the $|+\rangle$ or $|-\rangle$ states. She then transmits each prepared qubit to Bob through the quantum channel.

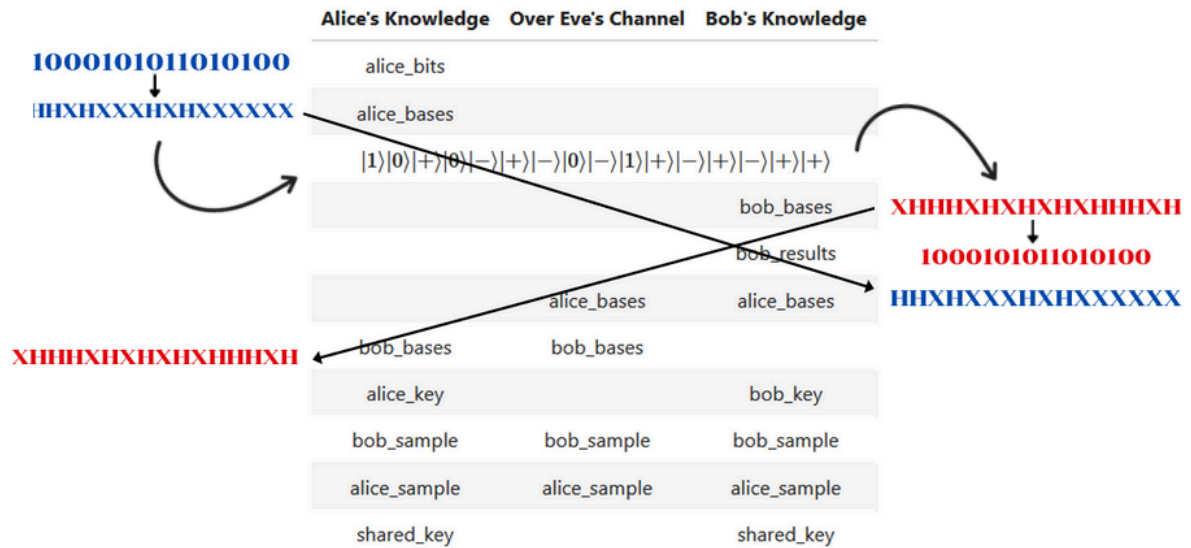
Quantum Key Distribution



Bob's Measurement. Bob independently generates his own random basis string of length n and measures each received qubit in his chosen basis. Applying H before measuring in Z if he chose the X basis, or applying X if he chose the Z basis. The critical observation is what happens depending on whether Alice and Bob's bases agree:

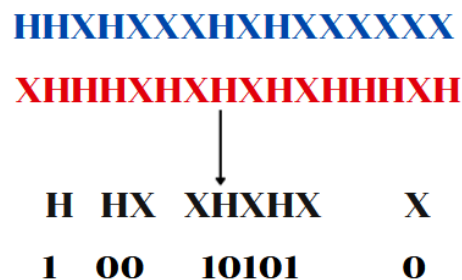


- **Bases match:** $XX=I$ and $HH=I$: the measurement undoes the encoding exactly, and Bob recovers Alice's original bit with certainty.
- **Bases differ:** Bob measures in an incompatible basis. The received state is not an eigenstate of his measurement operator, so the outcome is uniformly random, a 50/50 result of 0 or 1. Carrying no information about Alice's original bit.



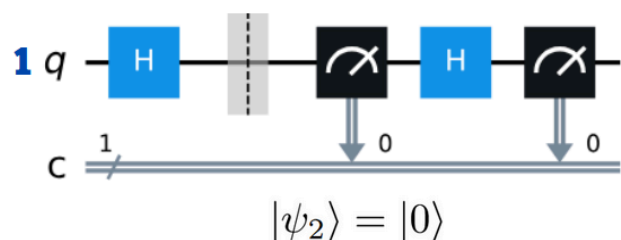
Sifting. After all qubits are transmitted and measured, Alice and Bob communicate over the classical channel to compare their basis choices for each position without revealing the actual bit values.

Alice and Bob kept only the key bits that has the same Basis



They discard all positions where their bases differed, retaining only the subset where they agree. This surviving subset is the sifted key, which in expectation retains approximately half the original bits.

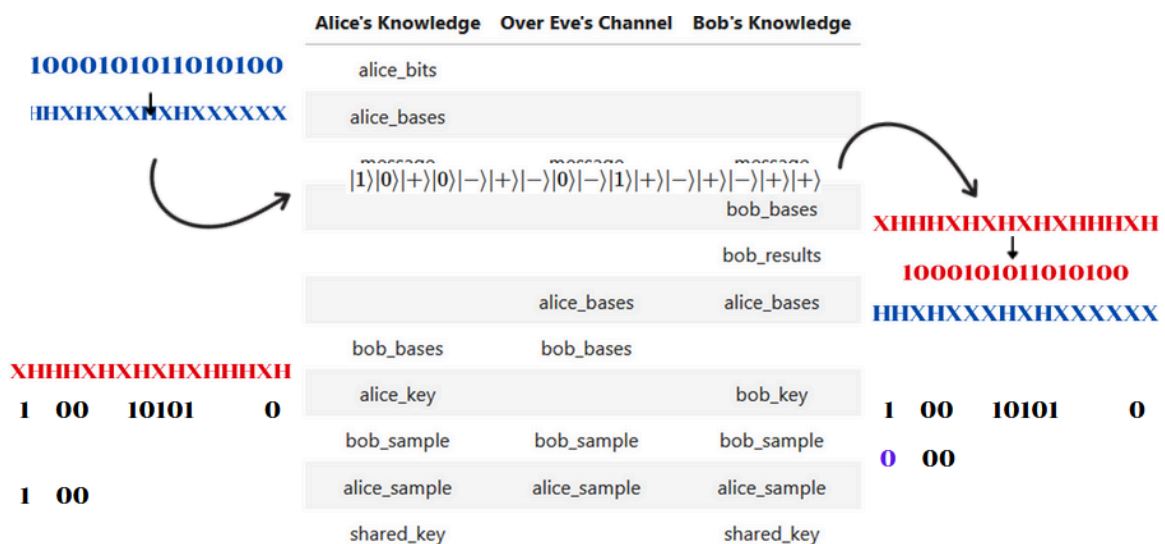
Eavesdropper Detection. If Eve intercepts a qubit on the quantum channel, she must measure it to extract information, but she



Eve's Interception (Evesdropping)

does not know Alice's basis choice. Half the time she will measure in the wrong basis, collapsing the qubit into a state inconsistent with Alice's original encoding. Even when she re-prepares and forwards a qubit to Bob, these disturbed states introduce errors into positions where Alice and Bob's bases agree positions that should be perfectly correlated. Alice and Bob detect this by publicly comparing a random sample of their sifted key bits over the classical channel. In the absence of eavesdropping, these bits should be identical. Any discrepancy elevates the QBER above the noise threshold, revealing Eve's presence and causing the session to be aborted.

$$|\psi_0\rangle = |1\rangle \longrightarrow |\psi_4\rangle = |0\rangle$$



Key Reconciliation and Privacy Amplification. If the QBER is below the threshold, Alice and Bob proceed with information reconciliation to correct residual errors from channel noise, followed by privacy amplification to eliminate any marginal information Eve may have gathered. The result is a final symmetric key shared exclusively between Alice and Bob, ready to seed the AES session cipher in the TLS layer.

Device-Independent Quantum Key Distribution

Standard QKD protocols such as BB84 rest on an implicit but critical assumption: that the devices used by Alice and Bob behave as specified. The security proofs assume that Alice's source genuinely produces quantum states, that Bob's measurement apparatus performs the declared quantum measurements, and that neither device leaks information to an adversary.

In practice this assumption is difficult to guarantee. A malicious device manufacturer, or an adversary who has compromised the hardware, could supply devices that appear to perform quantum operations but are in fact entirely classical, generating predetermined bit strings, simulating the expected behavior of a quantum channel while secretly transmitting the key to a third party. Under such a compromise, the security guarantees of BB84 and its variants collapse entirely regardless of how carefully the protocol is executed.

Device-Independent QKD (DI-QKD) addresses this threat by removing the need to trust the internal workings of any device. Instead of assuming that the devices are quantum, DI-QKD demands that the devices **prove** their quantum nature through their observable statistics alone.

The core insight is that certain correlations between measurement outcomes are impossible to produce classically. They can only arise if the devices are genuinely performing quantum operations on entangled states. If Alice and Bob's devices produce statistics that violate a classical bound, they have certified quantumness from the outside, without ever inspecting the hardware. This certification of quantumness is achieved through two complementary approaches, each exploiting a different manifestation of quantum behavior that has no classical analog.

Quantum non-locality (Entanglement)

**Via Bell's Inequality Violation
and**

Via Quantum Pseudo Telepathy

QKD CHSH (E91)

The CHSH game provides a concrete operational framework for certifying quantum behavior between two untrusted devices. In the game, a referee sends a random input bit $x \in \{0,1\}$ to Alice and a random input bit $y \in \{0,1\}$ to Bob. Alice and Bob must each respond with output bits a and b respectively, without communicating, satisfying the winning condition:

CHSH: Clauser-Horne-Shimony-Holt

(x, y)	win	lose
$(0, 0)$	$a = b$	$a \neq b$
$(0, 1)$	$a = b$	$a \neq b$
$(1, 0)$	$a = b$	$a \neq b$
$(1, 1)$	$a \neq b$	$a = b$

Classical

It is impossible to satisfy all four equations simultaneously, since summing them modulo 2 yields $0 = 1$. Therefore it is impossible to satisfy Condition

$$a \oplus b = x \wedge y$$

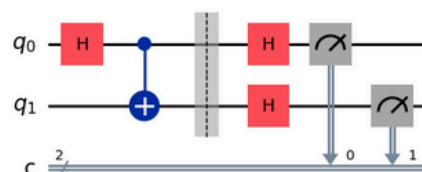
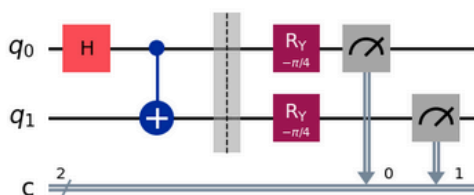
That is, their outputs must agree everywhere except when both inputs are 1, in which case their outputs must differ. A classical strategy, regardless of any pre-shared agreement, can satisfy this condition at most 75% of the time, this is the classical bound, provable by exhaustive enumeration of all deterministic strategies. Quantum mechanics allows Alice and Bob to exceed this bound by sharing an entangled state and choosing their measurements carefully, achieving a winning probability of $\cos^2(\pi/8) \approx 85\%$ which corresponds exactly to the Tsirelson bound.

Quantum Strategy. Alice and Bob share a maximally entangled pair $|\Phi^+\rangle = 1/2(|00\rangle + |11\rangle)$

Rather than responding with fixed classical outputs, each party applies a rotation $R_y(\theta)$ to their qubit and measures in the Z basis, where the angle θ is chosen based on their input. Alice randomly selects her measurement angle from $\{0^\circ, 45^\circ, 90^\circ\}$ and Bob from $\{45^\circ, 90^\circ, 135^\circ\}$. The asymmetric angle sets are chosen so that when Alice and Bob's angles are close, their measurement outcomes are strongly correlated, and the resulting correlations violate the CHSH inequality at the optimal level.

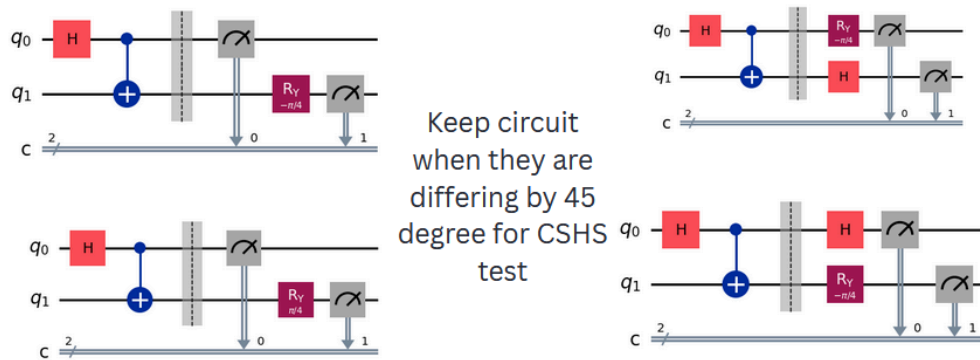
Key Generation. After each round of the game, Alice and Bob communicate their chosen angles over the classical channel — without revealing their measurement outcomes. The rounds are then partitioned into three categories:

- **Same angle:** Alice and Bob chose identical angles. Their measurements are perfectly correlated due to the $|\Phi^+\rangle$ entanglement, so their outcomes are guaranteed to agree. These rounds form the raw key — the shared bit is simply the common measurement outcome.

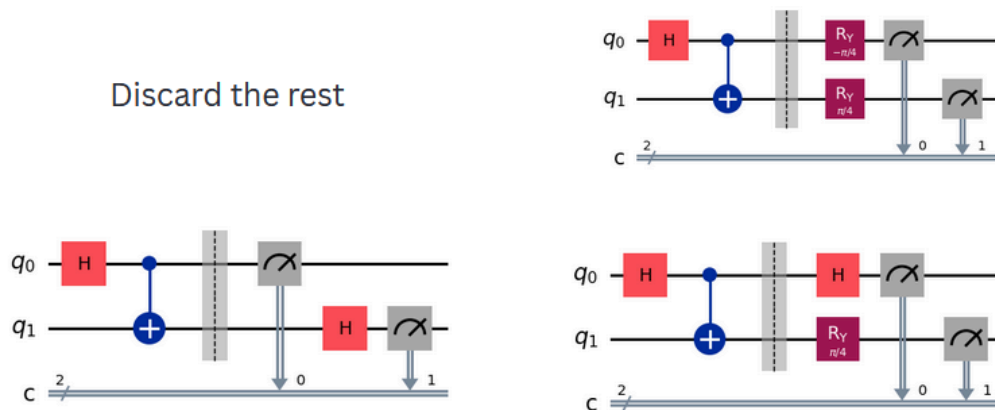


Keep circuit when they are in the same

- Angles differing by 45°:** The correlation between outcomes is $\cos^2(45^\circ/2) = \cos^2(22.5^\circ)$, which contributes to the 85% winning probability. These rounds are used to **estimate the CHSH winning rate** and certify that the devices are genuinely quantum. Alice and Bob accumulate these rounds and compute the empirical winning probability. If it approaches 85%, the Bell inequality is violated and the devices are certified as quantum. If it falls toward 75%, the devices are behaving classically potentially compromised and the session is aborted.



- Angles differing by 90° or more:** The correlation between outcomes is too weak to contribute usefully to either key generation or Bell testing. These rounds are discarded



entirely.

Device Certification and Security. The CHSH winning rate estimated from the 45°-difference rounds serves as the device integrity check. Unlike BB84 where security rests on trusting that the devices are quantum, here the Bell inequality violation is a self-certifying witness, no internal

knowledge of the devices is required. A classical device, regardless of how sophisticated, cannot consistently achieve a winning rate above 75%. A winning rate approaching 85% certifies not only that the devices are quantum but that the measurement outcomes used for key generation carry genuine quantum randomness that an adversary even one who built the devices cannot predict.

The key bits extracted from the same-angle rounds are then passed through information reconciliation and privacy amplification, exactly as in BB84, to produce the final secure key. The security guarantee now holds against device-level attacks that BB84 is entirely blind to.

```
--- Calculating Win Probabilities ---  
Inputs (x=0, y=0): Target (a XOR b) = 0. Win Probability = 0.8516  
Inputs (x=0, y=1): Target (a XOR b) = 0. Win Probability = 0.8516  
Inputs (x=1, y=0): Target (a XOR b) = 0. Win Probability = 0.8496  
Inputs (x=1, y=1): Target (a XOR b) = 1. Win Probability = 0.8604  
  
--- Overall Performance ---  
Experimental Average Win Probability: 0.8533  
Theoretical Quantum Win Probability: 0.8536  
Classical Limit Win Probability:      0.7500
```

Classical Win rate: 75% Quantum Win rate: 85%

CHSH: Clauser-Horne-Shimony-Holt

```
Circuit 1 (CHSH) win rate: 83.50%  
Circuit 2 (Discarded) results are not used.  
Circuit 3 (CHSH) win rate: 85.50%  
Circuit 4 (KeyGen) results are the same: True  
Circuit 5 (CHSH) win rate: 88.00%  
Circuit 6 (Discarded) results are not used.  
Circuit 7 (CHSH) win rate: 86.00%  
Circuit 8 (KeyGen) results are the same: True  
Circuit 9 (Discarded) results are not used.  
  
Total win rate: 85.75%
```

QKD Mermin's Magic Square

The second approach to device-independent key distribution replaces the statistical Bell witness of E91 with a stronger certification based on quantum pseudo-telepathy. The Mermin magic square game has a perfect quantum winning strategy but no perfect classical strategy. This makes it an attractive primitive for DI-QKD: rather than accumulating a winning rate over many rounds and checking whether it exceeds a statistical threshold, a single consistent pattern of perfect wins certifies quantum behavior outright.

Magic square game

x_1	x_2	x_3
x_4	x_5	x_6
x_7	x_8	x_9

$$x_i \in \{0, 1\}$$

Row sums are even.

Column sums are odd.

Referee assigns Alice (A) a row and Bob (B) a column. A and B return triplets of bits that follow the sum rules and also agree on the intersecting square.

The Game. The Mermin magic square is a 3×3 grid of two-qubit Pauli observables. A referee assigns Alice a random row $r \in \{1, 2, 3\}$ and Bob a random column $c \in \{1, 2, 3\}$. Alice must fill in her entire row with binary values $\{0, 1\}$ satisfying the row parity constraint. The sum of her three entries must be even parity. Bob must fill in his entire column satisfying the column parity constraint, the product of his three entries must be odd parity. The winning condition requires additionally that Alice and Bob agree on the value they assign to the intersecting cell (r, c) .

Classical

x_1	x_2	x_3
x_4	x_5	x_6
x_7	x_8	x_9

Sum of the entire table

=

even

odd

Contradiction

There does not exist a way to fill the table that wins 100%

Classically, no pre-agreed strategy can satisfy both parity constraints simultaneously across all rows and columns. The contradiction is algebraic: summing all row products gives even while summing all column gives odd, yet both operations sum the same nine cells, an impossibility for any fixed classical assignment. The best classical winning rate is therefore $8/9 \approx 89\%$, with exactly one input combination always failing.

Quantum

Prepare 4 qubits each in a Φ^+ entangle state

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

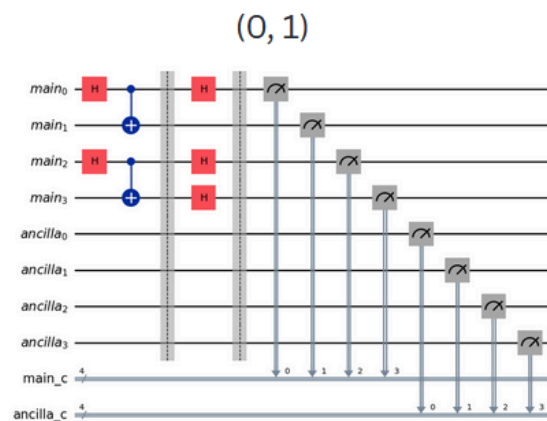
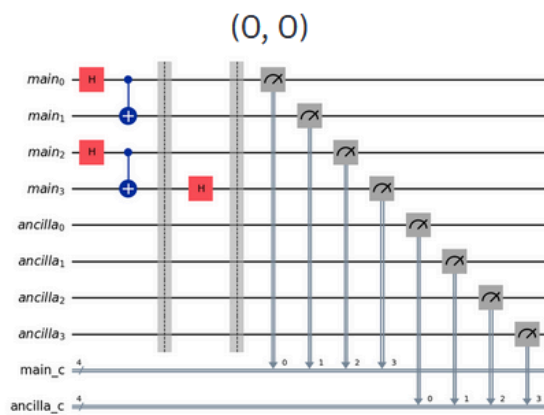
Quantum mechanically, Alice and Bob share two maximally entangled pairs $|\Phi^+\rangle \otimes |\Phi^+\rangle$ and perform joint measurements on their respective qubits corresponding to the Pauli observables of their assigned row or column. The operator identity $O_1 O_2 O_3 = +I$ for rows and $O_1 O_2 O_3 = -I$ for columns guarantees that the parity constraints are satisfied on every single shot, not just on average. As this is a state-independent algebraic fact baked into the structure of the Pauli group. The entanglement further guarantees that Alice and Bob's outcomes at the intersecting cell are always identical. The quantum winning rate is therefore exactly **100%**.

B_{11}	B_{12}	B_{13}	1
B_{21}	B_{22}	B_{23}	1
B_{31}	B_{32}	B_{33}	1
-1	-1	-1	

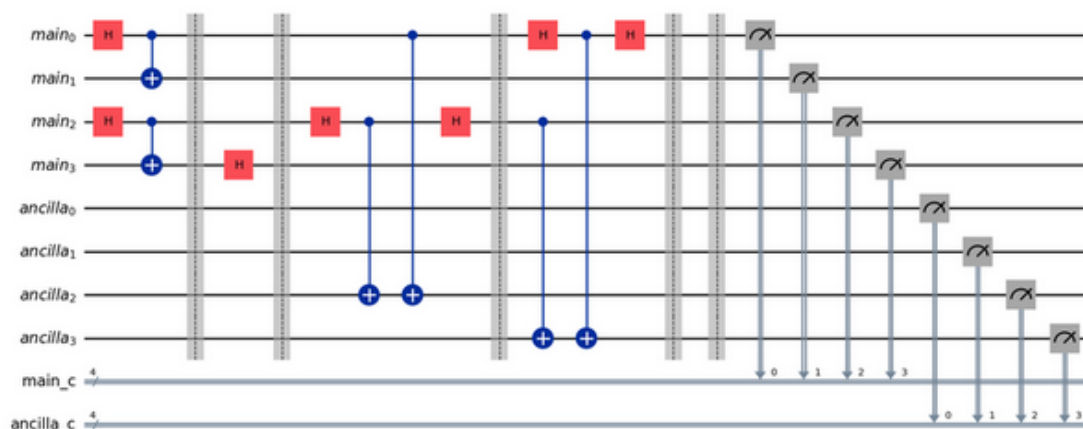
$\sigma_0 \otimes \sigma_z$	$\sigma_x \otimes \sigma_0$	$\sigma_x \otimes \sigma_z$
$\sigma_z \otimes \sigma_0$	$\sigma_0 \otimes \sigma_x$	$\sigma_z \otimes \sigma_x$
$-\sigma_z \otimes \sigma_z$	$-\sigma_x \otimes \sigma_x$	$-\sigma_y \otimes \sigma_y$

Key Generation. Alice and Bob play many rounds of the game. After each round they announce their assigned row and column over the classical channel, without revealing their full outcome strings. From the intersection cell (r,c) they each extract a single bit

Their agreed outcome at that cell serves as the key bit for that round. Since the quantum strategy guarantees perfect agreement at the intersection, these bits are identical for Alice and Bob without any reconciliation. Across many rounds the collection of intersection bits forms the raw key.



(0, 2)



Device Certification. After a sufficient number of rounds, Alice and Bob publicly compare a sample of their full outcome strings, not just the intersection bits. To verify that the parity constraints are satisfied and that their intersection values agree. If the observed winning rate approaches 100%, the devices are certified as quantum, no classical device can sustain a perfect win rate given the algebraic impossibility of the classical strategy. If the winning rate drops toward 8/9 or below, the devices are behaving classically or the channel has been compromised, and the session is immediately aborted.

Magic square game

0	1	1
0		
1		

100% Win rate

	1	
1	1	0
	1	

Comparison with E91. The Mermin magic square approach differs from E91 in a fundamental way. E91 detects eavesdropping and certifies quantumness statistically, the CHSH winning rate must be accumulated over many rounds before a judgment can be made, and the threshold of

85% versus 75% leaves a gap that requires careful statistical analysis. The magic square certification is categorical a classical device either fails or it does not, and perfect winning is logically impossible classically regardless of how many rounds are played. This makes the magic square a stronger and more direct witness of quantum behavior, though it requires two entangled pairs per round rather than one.

CODE: <https://github.com/YeYethePooh/QKD.git>